Resilience Academy

SDI System Installation & Customization with CRD and KoboToolbox

Session 1.2: Introduction to Software & Applications

Tools and Technologies for SDI Development

Masoud Hamad (SUZA) | **Msilikale Msilanga** (UTU)

Day 1 | Lecture 2 (45 min) | crd.resilienceacademy.ac.tz

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Session Overview

Goal: Know every tool in the SDI toolkit and why it is needed

By end of this session you can explain the role of each software component and verify hardware readiness.

Topics

- Docker & Docker Compose
- Git version control
- VS Code, Python 3, QGIS
- Postman & pgAdmin
- Hardware requirements

Activities

- Software checklist handout
- Instructor demo of each tool
- Verify your machine meets requirements

Docker & Docker Compose

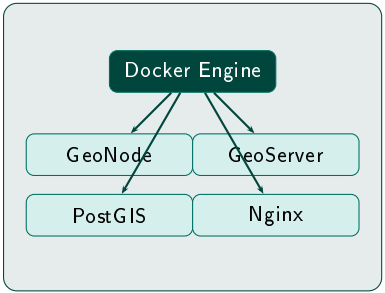
What is Docker?

- A **containerisation platform** — packages apps with all dependencies
- Containers are lightweight, portable, and reproducible
- No more "it works on my machine" problems
- CRD runs as a **multi-container stack** via Docker

Docker Compose

- Defines multi-container apps in a single `docker-compose.yml`
- One command: `docker compose up -d`

Host Machine (Ubuntu/WSL2)



Docker manages isolated containers

Git — Version Control System

Why Git?

- Tracks **every change** to your code
- Enables collaboration across teams
- CRD project lives on **GitHub** — you clone it to install
- Essential for reproducibility and rollback

Core Workflow

Key Git Commands

Command	Purpose
`git clone`	Download repository
`git status`	See what changed
`git add .`	Stage all changes
`git commit`	Save snapshot
`git push`	Upload to remote
`git pull`	Sync from remote
`git log`	View history

Day 2 Preview

```
git clone https://github.com/
GeoNode/geonode-project.git
```

VS Code, Python 3 & GeoPandas

VS Code

- Free, open-source code editor by Microsoft
- Integrated terminal, Git, extensions
- Used throughout this training for editing configs, Python scripts, and Docker files
- Extensions: Python, Docker, Remote-SSH

Python 3 + GeoPandas

- Python 3 — scripting, automation, data analysis
- **GeoPandas** — geospatial data in DataFrames
- Used in Day 4: fetch CRD WFS data, analyse Mwanza businesses, generate charts
- Also: `requests` for API calls, `matplotlib` for plots

Verification Commands

```
code -version | python3 -version | pip3 install geopandas
```

QGIS, Postman & pgAdmin

QGIS

- Free & open-source GIS
- Connects to CRD via WMS/WFS
- Visualise, query, and style Mwanza and Tanga layers
- Day 4: load CRD layers directly in QGIS

Postman

- API testing tool
- Send HTTP requests to CRD REST API
- Inspect JSON responses visually
- Day 4: explore `/api/v2/datasets/`

pgAdmin

- PostgreSQL GUI manager
- Browse CRD database tables
- Run SQL queries on PostGIS layers
- Day 2: inspect database after CRD install

Download All Before Day 2

QGIS: `qgis.org` | Postman: `postman.com` | pgAdmin: `pgadmin.org`

Hardware Requirements & Software Checklist

Minimum Hardware

Component	Minimum
RAM	8 GB
Disk space	20 GB free
OS	Ubuntu 22.04 LTS
Alternative	Windows + WSL2
CPU	2+ cores recommended
Network	Internet access required

Why Ubuntu 22.04?

All SDI software (GeoNode, KoboToolbox, GeoServer) is tested and documented for Ubuntu. WSL2 provides Ubuntu on Windows.

Software Checklist

- ☐ Docker & Docker Compose
- ☐ Git
- ☐ VS Code
- ☐ Python 3 + pip
- ☐ GeoPandas, requests, matplotlib
- ☐ QGIS
- ☐ Postman
- ☐ pgAdmin
- ☐ Web browser (Chrome/Firefox)

Practical 1.1 (afternoon) covers full installation step by step

The Complete SDI Software Stack

Client Layer: Browser, QGIS, Postman, KoboCollect

↓

Web Server: Nginx (reverse proxy, SSL termination)

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Application: GeoNode (Django) + GeoServer (Java) + KoboToolbox

↓

Database: PostgreSQL + PostGIS + MongoDB (Kobo)

↓

Infrastructure: Docker Compose + Git + Ubuntu 22.04

Key Insight

Every layer in this stack will be installed and configured during this 4-day training.

Session 1.2 Summary

Key Takeaways

- ✓ Docker containerises the entire CRD stack
- ✓ Docker Compose orchestrates multi-container apps
- ✓ Git tracks code and enables collaboration
- ✓ VS Code + Python + GeoPandas for analysis
- ✓ QGIS, Postman, pgAdmin for validation
- ✓ 8GB RAM + 20GB disk + Ubuntu 22.04

Next: Session 1.3

Introduction to Server Environments

- Local vs VM vs Cloud/VPS
- IP addresses, ports, SSH, DNS
- Common ports for SDI services
- Ubuntu 22.04 LTS setup

Questions?

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